

Eclipse Foundation invites tech leaders for collaboration on software-defined vehicles

The Eclipse Foundation, along with multiple industry leaders including Bosch, Microsoft and others, has announced an open invitation for technology leaders to help define a new working group focused specifically on the software-defined vehicle.

The ultimate goal will be the creation of a vendor-agnostic, open source ecosystem with a vivid, contributing community focused on building the foundation for a new era in automotive software development. This announcement serves as a “call to action” for all interested parties to join this initiative and help shape the future of mobility.

Today, next-generation vehicle developers are turning to software-based solutions for their new designs. The foundation believes this will lead to an open source revolution that results in software-defined vehicles. These vehicles will enable vehicle manufacturers as well as automotive suppliers to put software at the very centre of vehicle development, with hardware considerations to follow.

“We’re very excited to develop this new effort here at the Eclipse Foundation. Although we have extensive roots with the automotive community, a project of this scope and scale has never been attempted before,” said Mike Milinkovich, executive director of the Eclipse Foundation. “This initiative enables participants to get in at the ‘ground level’ and ensure they each have an equal voice in this project.”

To support the transformation to software-defined vehicles, major players from the technology industry as well as the automotive industry are being encouraged to collaboratively develop an open source in-vehicle application runtime stack, cloud-based vehicle operations, as well as highly integrated development toolchains.

The LFX Security module now includes automatic scanning for secrets-in-code and non-inclusive language, adding to its existing comprehensive automated vulnerability detection capabilities. Software security firm BluBracket has contributed this functionality to open source software projects under LFX as part of its mission of making software safer and more secure. This functionality builds on contributions from Snyk, making LFX a leading vulnerability detection platform for the open source community.



LFX Security now includes vulnerabilities detection, code secrets and non-exclusive language. LFX tracks how many known vulnerabilities have been found in open source projects, identifies if those vulnerabilities have

been fixed in code commits, and then reports on the number of fixes per project through an intuitive dashboard.

It detects secrets-in-code such as passwords, credentials, keys and access tokens, both pre- and post-commit. It also detects non-inclusive language used in project code, which is a barrier in creating a welcoming and inclusive community.

Microsoft’s Power Fx is now open source

Microsoft’s low-code programming language Power Fx has now been open sourced under an MIT licence on GitHub repository. The tech giant has open sourced the documentation of the language with plans to open source the actual source code by 2021 end.

Microsoft officially announced its Power Fx, an open source formula language for low code that’s based on Microsoft Excel, in March. This language is said to allow customers of the Power Platform to build and customise application logic. Using formulas that are already familiar to hundreds of millions of users, Power Fx allows a broad range of people to bring skills they



already know to low code solutions. It becomes a common ground for business users and professional developers alike to express logic and solve problems.

Power Fx is said to have the tools a professional expects, including the ability to directly edit apps in text editors like Visual Studio Code and use source control. This enables developers to go faster and find common ground with millions of makers.

Power Fx doesn’t just share the same syntax and functions as Excel, it also behaves in a familiar way. Like Excel, formulas are declarative and recalculate instantly just as a spreadsheet does. Makers have the advantage of telling the app what they want it to do without having to describe the how or when—Power Fx does that for you, freeing developers from the tedious task of keeping variables and data tables up to date manually.